

Ma-Ke Yuan (袁马轲)

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Education

Doctor of Science in Physics

Sep. 2023 – Jun. 2028 (anticipated)

Fudan University, Department of Physics

Advisor: Yang Zhou

Bachelor of Science in Physics (with Honours)

Sep. 2019 – Jun. 2023

Fudan University, Department of Physics

GPA: 3.83 / 4.0 (Rank: 3 / 85)

Research Interests

Quantum Field Theory with Defects; Multipartite and Mixed-State Quantum Correlations;
Topological Quantum Field Theory; Black Holes

Preprints and Publications

- Zi-Xiao Huang, **Ma-Ke Yuan** and Yang Zhou, to appear, [arXiv:2604.xxxxx](#).
- **Ma-Ke Yuan**, Mingyi Li and Yang Zhou, *Multi-entropy from Linking in Chern-Simons Theory*, [arXiv:2510.18408](#).
- Zi-Xiao Huang, **Ma-Ke Yuan** and Yang Zhou, *From Weyl Anomaly to Defect Supersymmetric Rényi Entropy and Casimir Energy*, [arXiv:2501.09498](#).
- **Ma-Ke Yuan**, Mingyi Li and Yang Zhou, *Reflected Multientropy and Its Holographic Dual*, *Phys. Rev. Lett.* **135** (2025) 091604, [arXiv:2410.08546](#).
- **Ma-Ke Yuan** and Yang Zhou, *Rényi Entropy with Surface Defects in Six Dimensions*, *JHEP* **03** (2024) 031, [arXiv:2310.02096](#).
- **Ma-Ke Yuan** and Yang Zhou, *Defect Localized Entropy: Renormalization Group and Holography*, *Nucl. Phys. B* **994** (2023) 116301, [arXiv:2209.08835](#).
- Yilu Shao, **Ma-Ke Yuan** and Yang Zhou, *Entanglement Negativity and Defect Extremal Surface*, *SciPost Phys. Core* **7** (2024) 027, [arXiv:2206.05951](#).
- Tianyi Li, **Ma-Ke Yuan** and Yang Zhou, *Defect Extremal Surface for Reflected Entropy*, *JHEP* **01** (2022) 018, [arXiv:2108.08544](#).

Conference Presentations

- *Multi-entropy from Linking in Chern-Simons Theory*, Short Talk, Recent Progress in Quantum Field Theory and Gravity, Fudan University, Dec. 17–19, 2025.

- *Reflected Multi-entropy*, Oral Presentation (first Prize), The 12th PFUNT PhD Student Forum, Nanjing University, Dec. 5–7, 2025.
- *Rényi Entropy with Surface Defects in Six Dimensions*, Gong Show, The Physics and Mathematics of Boundaries, Impurities, and Defects, Isaac Newton Institute for Mathematical Sciences, Cambridge, Sep. 8–12, 2025.
- *Rényi Entropy with Surface Defects in Six Dimensions*, Short Talk, New advancements on defects and their applications, Yukawa Institute for Theoretical Physics, Kyoto University, Jul. 14–18, 2025. <https://indico.yukawa.kyoto-u.ac.jp/event/63/contributions/1084/>
- *Rényi entropy with surface defects in 6d (2,0) theories*, Oral Presentation (second prize) & Poster (first prize), The 11th PFUNT PhD Student Forum, University of Science and Technology of China, Dec. 5–7, 2024.
- *Rényi Entropy with Surface Defects in 6d*, Gong Show & Poster, School of Frontiers in Hydrodynamics, Effective Field Theory and Holographic Duality, Jilin University, Jul. 7–22, 2024.
- *Defect Localized Entropy and Defect C-Theorem*, Oral Presentation (first prize), The 10th PFUNT PhD Student Forum, Fudan University, Dec. 8–10, 2023.
- *Defect Localized Entropy: Renormalization Group and Holography*, Poster, International Congress of Basic Science, Yanqi Lake Beijing Institute of Mathematical Sciences and Applications, Jul. 16–28, 2023.

Awards

National Scholarship for Doctoral Students (2025)
 National Scholarship for Master Students (2023, 2024)
 Bachelor of Science with Honours in Physics (2023)
 Top Students Award in Physics (2023)

Skills

Mathematica; Python; Julia; C